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Pavan Kumar

Full-Stack AI Engineer

AI Engineer with 4+ years of engineering experience, including 2 years shipping GenAI features end-to-end, from prompt engineering and RAG pipeline design through to cloud deployment on AWS/Azure. Built and maintained LLM-powered products that reduced manual tasks by 70% and increased accuracy by 30%. Comfortable owning the full ML life cycle in fast-moving teams, with strong instincts for trade-offs between model quality, cost and speed. Excited to help a product team move from an AI prototype to a working product at pace.

SKILLS

AI, ML & Cloud	PyTorch, scikit-learn, LangChain, LangGraph, LangSmith, GoogleADK, Optuna, AWS, Azure
Tools & Technical languages	Git, Claude, MongoDB, Python, Linux, SQL, Node.js, React, TypeScript
Languages	English (Fluent), German (A2), Hindi (Native)
MLOps / AIOps	Docker, Kubernetes, GitHub Actions CI/CD, Terraform, Mlflow, Kubeflow, ArgoCD and Prometheus

EDUCATION

MASTER'S IN MATHEMATICS IN SCIENCE AND ENGINEERING (MAJOR IN AI) **April 2022 – Jul 2026**
Technical University of Munich (TUM) Munich, Germany

Coursework: Machine Learning, Deep Learning, Computer Vision, Foundations of LLMs, Numerics and Non-Linear Optimization.

B.TECH ENGINEERING (MAJOR IN MATHEMATICS) **June 2013 – May 2017**
Vellore Institute of Technology Vellore, India

Coursework: Probability & Statistics, Differential & Difference Equations, Computer Programming C and Complex Variables & PDEs.

TECHNICAL EXPERIENCE

AIOPS ENGINEER, INTERN **Sep 2025 – Feb 2026**
Munich Re Munich, Germany

- Architected a multi-step credit-control workflow in Microsoft Power Automate, integrating LLM tool-calling and human-in-the-loop approvals to automate checks and route decisions to responsible stakeholders, replacing a fully manual end-to-end process.
- Deployed an agentic RAG backend with FastAPI, Docker and AWS EKS, using GitHub Actions for CI/CD and LangSmith for monitoring.

AI ENGINEER, WORKING STUDENT **Jun 2025 – Aug 2025**
MERMEC Group Munich, Germany

- Built a multi-agent system with LangGraph (Deepagents) and hybrid RAG on internal technical documentation, for engineers.
- Containerized the app with Docker, ran it on Kubernetes on AWS and used GitLab CI/CD to ship the application to production.

DATA ENGINEER, WORKING STUDENT **Jun 2024 – Sep 2024**
BMW Group Munich, Germany

- Built Azure Databricks ETL pipelines for vehicle sensor data, with SQL analysis and Power BI KPI reporting.

ML ENGINEER, INTERN, FULL-TIME **Sep 2023 – Apr 2024**
ZEISS Group, Corporate Research & Technology Oberkochen, Germany

- Implemented polynomial regression from scratch in PyTorch to interpolate 3D printer data.
- Tuned model architectures with the tool Optuna and deployed the final model to production on AWS.

DIGITAL PLATFORM & AUTOMATION, INTERN **Aug 2022 – Jan 2023**
Infineon Technologies Munich, Germany

- Automated training data processing with Python/Pandas and streamlined digital content deployment workflows on Infineon's platform.

ML ENGINEER, INTERN **Jun 2023 – Aug 2023**
MaxPlanck Garching, Munich

- Built Scikit-learn pipelines for spectral image preprocessing and classification, benchmarking Random Forest/XGBoost against CNNs.

PROJECT MANAGER (DEVOPS ENGINEER), FULL-TIME **Mar 2019 – Sep 2020**
HomeLane Pune, India

- Developed HomeLane's internal full-stack platform and made it scalable and reliable on AWS, using Terraform for infrastructure, CI/CD for automated deployments and Prometheus and Grafana for monitoring.

PROJECTS

- **Master's Thesis (TUM):** Developed a Neural Process framework for spatio-temporal air-quality forecasting (AOD and PM2.5) from MODIS satellite data, leveraging Bayesian Neural Networks, Gaussian Processes, CNNs, cKDTree and KNNs.
- **Personal:** Production-grade supply-chain agent using LangGraph, RAG, tool calling, safety guardrails, observability and CI/CD.

ACHIEVEMENTS AND HOBBIES

4th place in the TUM.ai Makeathon Challenge among 40+ teams; Calisthenics, cooking and volunteering.